

CARANSEBES, Romania

WMO: 152920

Lat: 45.420N

Lon: 22.253E

Elev: 866

StdP: 14.24

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.2	16.9	6.9	8.1	15.3	11.7	10.3	19.4	26.2	34.5	22.7	35.6	2.8	150	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.7	91.6	69.8	88.5	69.7	85.4	68.8	73.5	85.8	71.7	84.0	70.0	82.0	6.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.5	111.9	80.2	67.7	105.0	77.1	66.1	99.4	75.0	37.5	85.7	35.9	84.1	34.5	82.0	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.3	16.9	14.2	DB	4.2	95.3	6.3	3.9	-0.3	98.1	-4.0	100.4	-7.6	102.6	-12.2	105.4	(4)
(5)			WB	3.8	76.3	6.1	2.5	-0.6	78.1	-4.2	79.6	-7.7	81.0	-12.1	82.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.8	32.8	36.4	43.7	53.7	61.3	67.2	70.5	70.9	62.3	53.6	45.1	35.1		
	DBStd	15.34	8.77	9.07	8.26	7.18	6.45	6.17	5.70	5.90	6.91	7.86	8.90	8.61		
	HDD50	1911	534	386	228	43	3	0	0	0	1	55	196	465		
	HDD65	5045	998	802	661	342	150	48	16	15	131	359	596	927		
	CDD50	2929	1	4	32	155	353	515	635	646	369	166	50	3		
	CDD65	590	0	0	0	4	35	113	186	197	49	6	0	0		
	CDH74	6321	0	0	1	65	386	1225	1998	2094	494	57	1	0		
	CDH80	2296	0	0	0	8	63	400	801	881	138	5	0	0		
(14) Wind	WSAvg	4.8	5.0	5.5	5.6	5.5	4.4	4.0	3.8	3.9	4.2	5.0	6.1	4.8		
(15) Precipitation	PrecAvg	34.7	1.9	2.0	2.0	2.9	3.3	4.5	4.2	3.7	3.5	2.4	2.2	2.2		
	PrecMax	51.2	3.5	6.5	3.8	7.4	7.6	8.0	7.6	10.1	8.3	5.3	5.9	5.0		
	PrecMin	24.1	0.6	0.3	0.4	0.4	1.2	0.9	1.5	0.6	0.7	0.2	0.5	0.6		
	PrecStd	7.2	0.8	1.4	0.9	1.6	1.5	1.9	1.7	2.4	1.8	1.4	1.3	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.8	62.8	70.2	80.3	85.1	91.4	96.2	96.8	89.7	79.2	69.9	58.3		
		MCWB	47.9	50.8	53.3	60.6	66.4	70.9	69.1	70.2	66.3	62.1	55.8	49.2		
	2%	DB	51.8	56.5	65.2	75.1	81.5	88.0	91.3	92.4	83.8	74.9	64.7	53.8		
		MCWB	45.5	47.4	51.3	58.8	64.7	70.6	69.8	69.8	65.8	60.4	54.5	47.2		
	5%	DB	47.9	52.4	60.9	70.7	78.5	84.6	88.1	88.6	79.5	70.7	61.0	50.3		
		MCWB	43.3	45.2	49.2	55.7	63.4	69.2	70.0	69.5	64.0	58.9	52.2	45.2		
	10%	DB	44.3	48.7	57.1	66.9	75.1	81.4	84.5	84.9	75.5	66.3	57.5	46.8		
		MCWB	41.1	43.4	47.5	54.0	62.0	67.6	69.2	68.4	63.0	57.1	50.6	42.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.9	51.4	55.2	62.6	69.7	75.0	75.4	76.0	70.8	64.9	57.6	50.7		
		MCDB	55.3	60.1	67.2	77.5	81.2	86.8	86.7	88.6	83.9	74.4	66.1	55.0		
	2%	WB	46.1	48.5	52.5	59.8	66.8	72.2	73.3	73.3	67.5	62.1	55.5	47.9		
		MCDB	50.8	55.0	62.3	72.6	77.5	84.5	86.8	86.8	77.9	71.6	62.7	53.0		
	5%	WB	43.6	46.1	50.4	57.4	64.8	70.2	71.7	71.1	65.6	60.0	53.6	45.5		
		MCDB	47.4	51.5	59.2	68.2	75.4	82.1	83.7	83.6	75.8	68.6	59.8	49.7		
	10%	WB	41.1	43.7	48.3	55.2	62.9	68.4	70.0	69.3	63.7	58.0	51.4	43.2		
		MCDB	44.1	48.3	55.8	64.1	73.0	79.6	81.7	81.3	73.5	65.4	57.0	46.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	13.7	17.3	19.3	20.4	21.1	22.2	22.7	19.7	18.3	14.0	11.0		
		MCDBR	17.1	19.3	23.0	25.4	25.8	26.3	28.0	28.4	26.3	24.3	18.5	16.1		
	5% WB	MCWBR	11.1	11.9	12.5	12.2	12.7	12.3	11.8	11.4	11.4	12.2	10.5	10.5		
		MCWBR	15.7	17.6	20.7	22.9	23.7	24.3	24.5	24.4	22.6	21.4	17.2	14.9		
(40) Clear-Sky Solar Irradiance	taub		0.329	0.359	0.403	0.463	0.449	0.439	0.454	0.468	0.431	0.388	0.355	0.329		
	taud		2.304	2.214	2.172	2.035	2.139	2.214	2.175	2.129	2.198	2.310	2.348	2.342		
	Ebn at Noon		241	254	259	252	261	264	258	248	247	242	229	228		
	Edh at Noon		26	34	41	51	47	44	45	46	39	31	25	23		
(44) All-Sky Solar Radiation	RadAvg		409	664	1029	1400	1673	1875	1858	1679	1189	826	490	336		
	RadStd		50	86	140	152	154	176	151	164	124	94	56	45		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.30	N/A	N/A	+2.03	N/A	N/A	N/A	-288	+173
(47) Regional (71 neighbors)		+1.22	N/A	N/A	N/A	N/A	N/A	-211	-324	+218

Nomenclature: See separate page